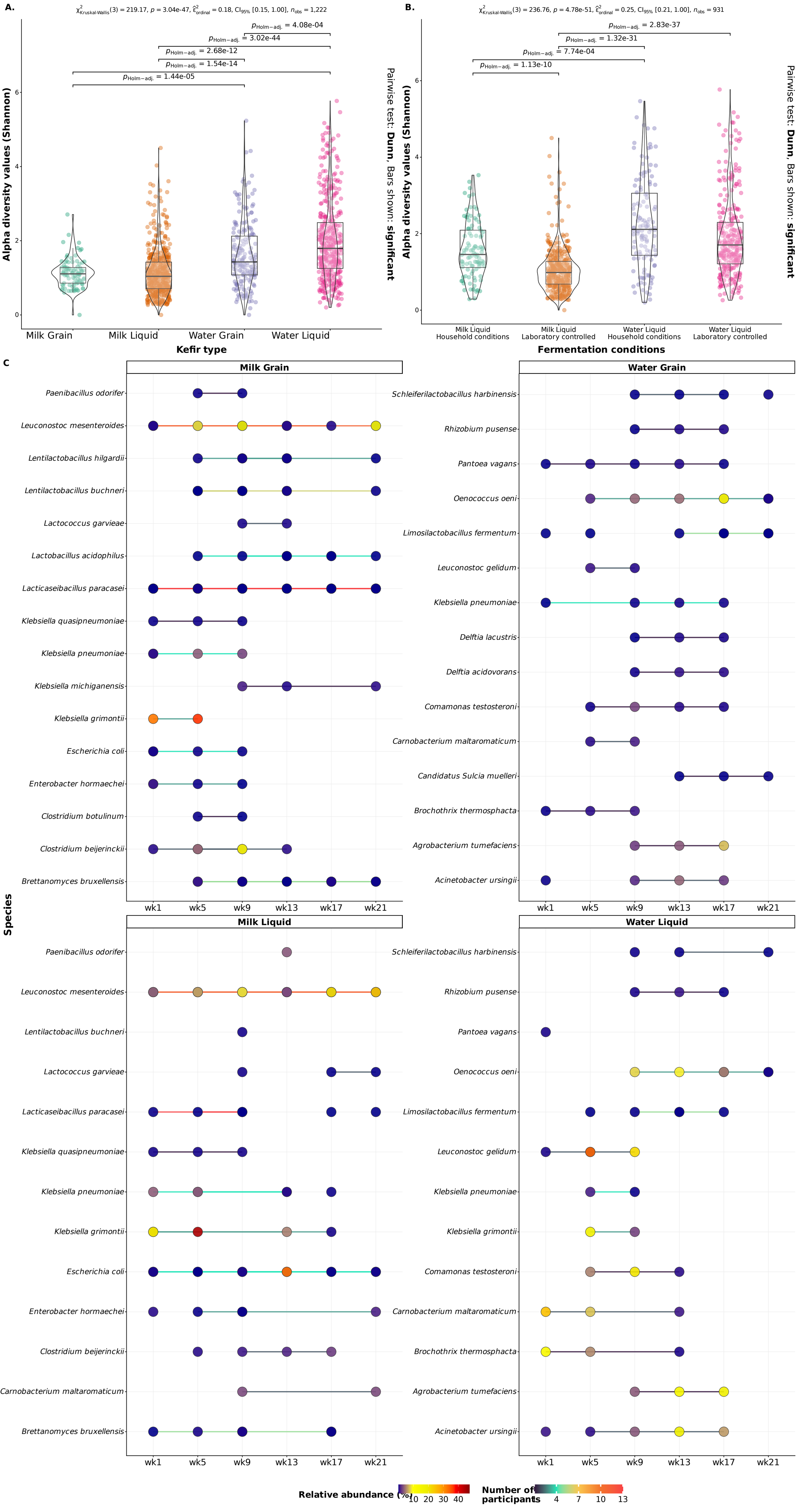

A.
$\chi^2_{Kruskal\text{-}Wallis}(3) = 219.17$, $p = 3.04e\text{-}47$, $\hat{\varepsilon}^2_{ordinal} = 0.18$, $CI_{95\%}$ [0.15, 1.00], $n_{obs} = 1,222$
$p_{Holm-adj.}$ = 4.08e-04
$p_{Holm-adj.}$ = 3.02e-44
$p_{Holm-adj.}$ = 2.68e-12
$p_{Holm-adj.}$ = 1.54e-14
$p_{Holm-adj.}$ = 1.44e-05
Alpha diversity values (Shannon)
Milk Grain
Milk Liquid
Water Grain
Water Liquid
Kefir type
Pairwise test: Dunn, Bars shown: significant
B.
$\chi^2_{Kruskal\text{-}Wallis}(3) = 236.76$, $p = 4.78e\text{-}51$, $\hat{\varepsilon}^2_{ordinal} = 0.25$, $CI_{95\%}$ [0.21, 1.00], $n_{obs} = 931$
$p_{Holm-adj.}$ = 2.83e-37
$p_{Holm-adj.}$ = 1.32e-31
$p_{Holm-adj.}$ = 7.74e-04
$p_{Holm-adj.}$ = 1.13e-10
Alpha diversity values (Shannon)
Milk Liquid Household conditions
Milk Liquid Laboratory controlled
Water Liquid Household conditions
Water Liquid Laboratory controlled
Fermentation conditions
Pairwise test: Dunn, Bars shown: significant
C
Milk Grain
Water Grain
Milk Liquid
Water Liquid
Species
Paenibacillus odorifer
Leuconostoc mesenteroides
Lentilactobacillus hilgardii
Lentilactobacillus buchneri
Lactococcus garvieae
Lactobacillus acidophilus
Lacticaseibacillus paracasei
Klebsiella quasipneumoniae
Klebsiella pneumoniae
Klebsiella michiganensis
Klebsiella grimontii
Escherichia coli
Enterobacter hormaechei
Clostridium botulinum
Clostridium beijerinckii
Brettanomyces bruxellensis
Schleiferilactobacillus harbinensis
Rhizobium pusense
Pantoea vagans
Oenococcus oeni
Limosilactobacillus fermentum
Leuconostoc gelidum
Delftia lacustris
Delftia acidovorans
Comamonas testosteroni
Carnobacterium maltaromaticum
Candidatus Sulcia muelleri
Brochothrix thermosphacta
Agrobacterium tumefaciens
Acinetobacter ursingii
wk1
wk5
wk9
wk13
wk17
wk21
Relative abundance (%)
10 20 30 40
Number of participants
1 4 7 10 13